Basic UNIX commands :

The basic Unix commands are:

echo

It displays a line text.

Echo the given string or strings to standard output.

Syntax:
echo [OPTION]... [STRING]...

Example:

echo "Good Morning"

This will display the following in the standard output

Good Morning

passwd command

The passwd command is used to change passwords for user accounts. A normal user may only change the password for his or her own account, while the superuser may change the password for any account.

Syntax : passwd [options] [LOGIN]

Options:

Short Option	Long Option	Option Description
-a	–all	This option can be used only with -S and causes show status for all users.
-d	–delete	Delete a users password (make it empty). This is a quick way to disable a password for an account. It will set the named account password-less.
-e	–expire	Immediately expire an accounts password. This in effect can force a user to change his/her password at the users next login.
-i	–inactive INACTIVE	This option is used to disable an account after the password has been expired for a number of days. After a user account has had an expired password for INACTIVE days, the user may no longer sign on to the account.
-k	–keep-tokens	Indicate password change should be performed only for expired authentication tokens (passwords). The user wishes to keep their non-expired tokens as before.
-l	–lock	Lock the password of the named account. This option disables a password by changing it to a value which matches no possible encrypted value (it adds a ´!´ at the beginning of the password). Note that this does not disable the account. The user may still be able to login using another authentication token (e.g. an SSH key). To disable the account, administrators should use usermod –expiredate 1 (this set the accounts expire date to Jan 2, 1970). Users with a locked password are not allowed to change their password.
-n	–mindays MIN_DAYS	Set the minimum number of days between password changes to MIN_DAYS. A value of zero for this field indicates that the user may change his/her password at any time.
-q	–quiet	Quiet mode.

Example:

```
$ passwd guest Changing
password for guest.
(current) UNIX password:
Enter new UNIX password:
Retype new UNIX password: passwd:
password updated successfully
```

Display the account status information using -S option

```
$ passwd -S  himanshu
himanshu P 03/12/2011 0 99999 7 -1
```

Delete an account password using the -d option

```
$ sudo passwd -d guest passwd: password
expiry information changed.
```

Immediately expire the account password using -e option

```
$ sudo passwd -e guest passwd: password
expiry information changed.
```

Perform actions quietly by using -q option

```
$sudo passwd -eq guest
```

uname command

Print certain system information.

Syntax : uname [OPTION]...

Options:

Tag	Description
-a, --all	print all information, in the following order, except omit **-p** and **-i** if unknown:
-s, --kernel-name	
	print the kernel name
-n, --nodename	
	print the network node hostname
-r, --kernel-release	
	print the kernel release
-v, --kernel-version	
	print the kernel version
-m, --machine	
	print the machine hardware name
-p, --processor	
	print the processor type or "unknown"
-i, --hardware-platform	
	print the hardware platform or "unknown"
-o, --operating-system	
	print the operating system
--help	display this help and exit
--version	
	output version information and exit

Examples: To show the name of the OS

uname command:

Used to get the information about the current system (the name, version and other details)

$uname

To get the kernel name, you can use -s parameter.

$ uname -s

To get the kernel release, -r option is used

$ uname -r

who command:

who command prints information about all users who are currently logged in.
Syntax:
who [OPTION]... [FILE] [am i]

-a, --all	Same as using the options -b -d --login -p -r -t -T -u.
-b, --boot	Display the time of the last system boot.
-d, --dead	Display dead processes.
-H, --heading	Print a line of column headings.
-l, --login	Print system login processes.
--lookup	Attempt to canonicalize hostnames via DNS.
-m	Only print information about the user and host associated with standard input (the terminal where the command was issued). This method adheres to the POSIX standard.
-p, --process	Print active processes spawned by init.
-q, --count	Displays all login names, and a count of all logged-on users.
-r, --runlevel	Print the current runlevel.
-s, --short	Print only name, line, and time fields. This is the default.

find command:

find is one of the powerful utility of Unix (or Linux). It can search the entire file-system to locate files and directories according to the specific search criteria. Other than searching files in can also perform actions(executing commands) on searched files.

Syntax:

$ find [option] [action]

Examples :

1 . File Type based Search:

>> The following command finds the file named Abc in current(.) directory and all its sub-directories

Tag	Description
-s, --silent, --quiet	
	print nothing, only return an exit status
--help	display this help and exit
--version	
	output version information and exit

2. Wild-Card based Search : Wild-card characters like * and ? can be used:

>> To find all files where filename starts with Abc the command will be:

[390119@InGhyUnix ~]$ find . -name "Abc*"
./Abc
./Dir1/Abc
./Dir1/Abc.php
./Abc.txt
./Abc.dat

3 . Source Location based Search:

>>The command find can search in multiple source location :

[390119@InGhyUnix ~]$ find Dir1 Dir2 -name "Abc.*"
Dir1/Abc.php
Dir2/Abc.htm

Searches for the file Abc with any extension only in directories Dir1 and Dir2

4 . Size Based Search : Using find files can be searched based on its size .
>> To find the files with size more than 10Mb the command will be

```
[390119@InGhyUnix ~]$ find . -size +10M

./Abc.dat
```

5 . Access/Modification-Time Based Search : Using find files can be searched based on the time of last access/modification :
>> To find the files which are accessed within last 3 days the command will be

```
[390119@InGhyUnix ~]$ find . -atime -3

./Abc
```

6 . Permission Based Search : Using find command files can be searched based on the access permission they have
>> To find the files having read write and execute permission only for the owner the command will be :

```
[390119@InGhyUnix ~]$ find . -perm 700
./Abc.txt
```

7 . Perform Action on the search result : Using exec option any action (command)
can be executed on the search result.
>> To remove all the files having extension .swp from the current directory the command will be :

```
[390119@InGhyUnix ~]$ find . -name "*.swp" -exec rm {} \;
```

bc command:

bc is the command used for basic mathematical calculations.

It can be used either a mathematical scripting language or as an interactive mathematical shell .

In interactive mode it accepts input from the terminal and providing calculations on requested arithmatic expression.

That means typing the command bc on a Unix command prompt and entering a mathematical expression, such as (1 + 4) * 2 , whereupon 10 will be the output

While used with files , bc starts by processing code from all the files listed on the command line in the order listed. After all files have been processed, bc reads from the standard input. All code is executed as it is read.

Syntax: bc [-hlwsqv] [long-options] [file ...]

Example:

$ echo "2+5" | bc

Output will be:

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tty command:
Prints the file name of the terminal connected in standard input.
Syntax : tty [OPTION]...

Tag	Description
-s, --silent, --quiet	
	print nothing, only return an exit status
--help	display this help and exit
--version	
	output version information and exit